SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

MTH316

Examination - January Semester 2026

Multivariable Calculus

Friday, 17 April 2026 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Define a function $f : \mathbb{R}^2 - \{(0,0)\} \longrightarrow \mathbb{R}$ by

$$f(x,y) = \frac{ky^2}{\sqrt{x^2+y^2}},$$

where k is a positive constant.

Question 1a(i)

Determine

$$\lim_{(x,y)\to(-1,1)} f(x,y),$$

if it exists. Or else, show that it does not exist.

(3 marks)

Question 1a(ii)

Evaluate

$$\lim_{(x,y)\to(0,0)} f(x,y),$$

if it exists. Or else, show that it does not exist.

(9 marks)

Question 1b

Define a function $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$ by

$$g(x,y) = \begin{cases} \dfrac{x^2-y^2}{cx-cy}, & \text{if } y \neq x \\ \dfrac{e^c}{e} - 1, & \text{otherwise} \end{cases}$$

where c is a non-zero constant. Determine the value of c for the function to be continuous at $(0,0)$.

(8 marks)

Question 2

During a manufacturing process, the pressure (in kPa) imposed by jets of air over the surface of a steel plate is defined by the function

$$P(x, y) = 2x^4 + y^2 - 8xy + 50.$$

Question 2a

At the point $(1, 0)$on the steel plate, calculate the directional derivative in the direction towards $(3, 3)$ and explain if the air pressure is increasing or decreasing at the point $(1, 0)$ towards $(3, 3)$.

(5 marks)

Question 2b

If the steel plate is not bounded, determine the critical points of the function P. Classify each critical point.

(9 marks)

Question 2c

Apply derivative tests to determine the absolute highest and lowest air pressures acting on the plate along the closed straight line segment from $(1, 0)$ to $(1, 5)$.

(6 marks)

Question 3

An object possesses the volumetric region E described by

$$E : \{(x, y, z) \mid -z \leq x \leq z,\ -z \leq y \leq z,\ 0 \leq z \leq 1.\}$$

Question 3a

Sketch the projections of the volumetric region E onto the Cartesian coordinate planes, namely the xy-plane, xz-plane and yz-plane. Label the axes intercepts in all three projections.

(6 marks)

Question 3b

Compute the volume of the volumetric region E by a suitable triple integral. Label the limits of integration clearly.

(9 marks)

Question 4

Define a force field by

$$\mathbf{F}(x, y) = \begin{bmatrix} -ky \\ kx \end{bmatrix},$$

where k is a constant. The figure below shows paths C_1 to C_3 that form a closed path where C_3 is a circular arc of radius R. The angles that C_1 and C_2 makes with the x-axis are 45° and -45° respectively.

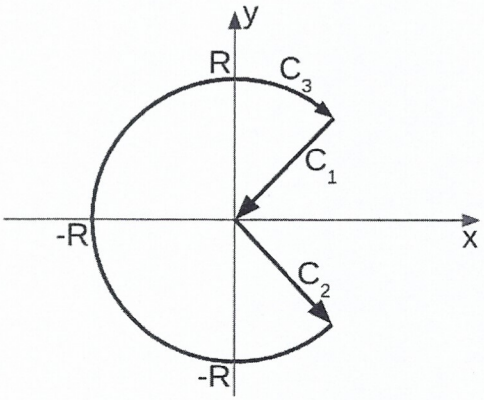

Figure Q4

Question 4a

Using line integrals, compute directly the work done by $\mathbf{F}(x, y)$ along paths C_1 and C_2.

(10 marks)

Question 4b

Describe the force field and hence explain the values of the work done found in Question 4a.

(5 marks)

Question 4c

Use Green's theorem to evaluate the work done by $\mathbf{F}(x, y)$ along path C_3.

(5 marks)

Question 5

A vector field $\mathbf{V}$ and a cylinder with a slanted cutoff are defined below, where all three surfaces of the cylinder are orientated with outward normal vectors. S_1 is the slanted cutoff surface of the cylinder that is defined by $y = 2 - z$. The cylinder has a radius of one.

$$\mathbf{V}(x, y, z) = \begin{bmatrix} 2x \\ 7y \\ 2z \end{bmatrix}.$$

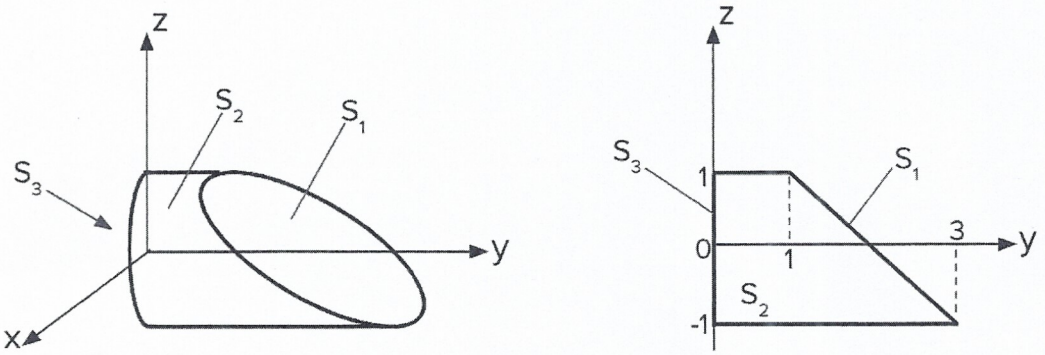

Figure Q5

Question 5a

Define a surface integral and compute the flux of $\mathbf{V}(x, y, z)$ through surface S_1.

(10 marks)

Question 5b

Without integration, state the flux of $\mathbf{V}(x, y, z)$ through surface S_3 and explain its value.

(5 marks)

Question 5c

Use the divergence theorem to evaluate the flux of $\mathbf{V}(x, y, z)$ through S_2.

(10 marks)

----- END OF PAPER -----